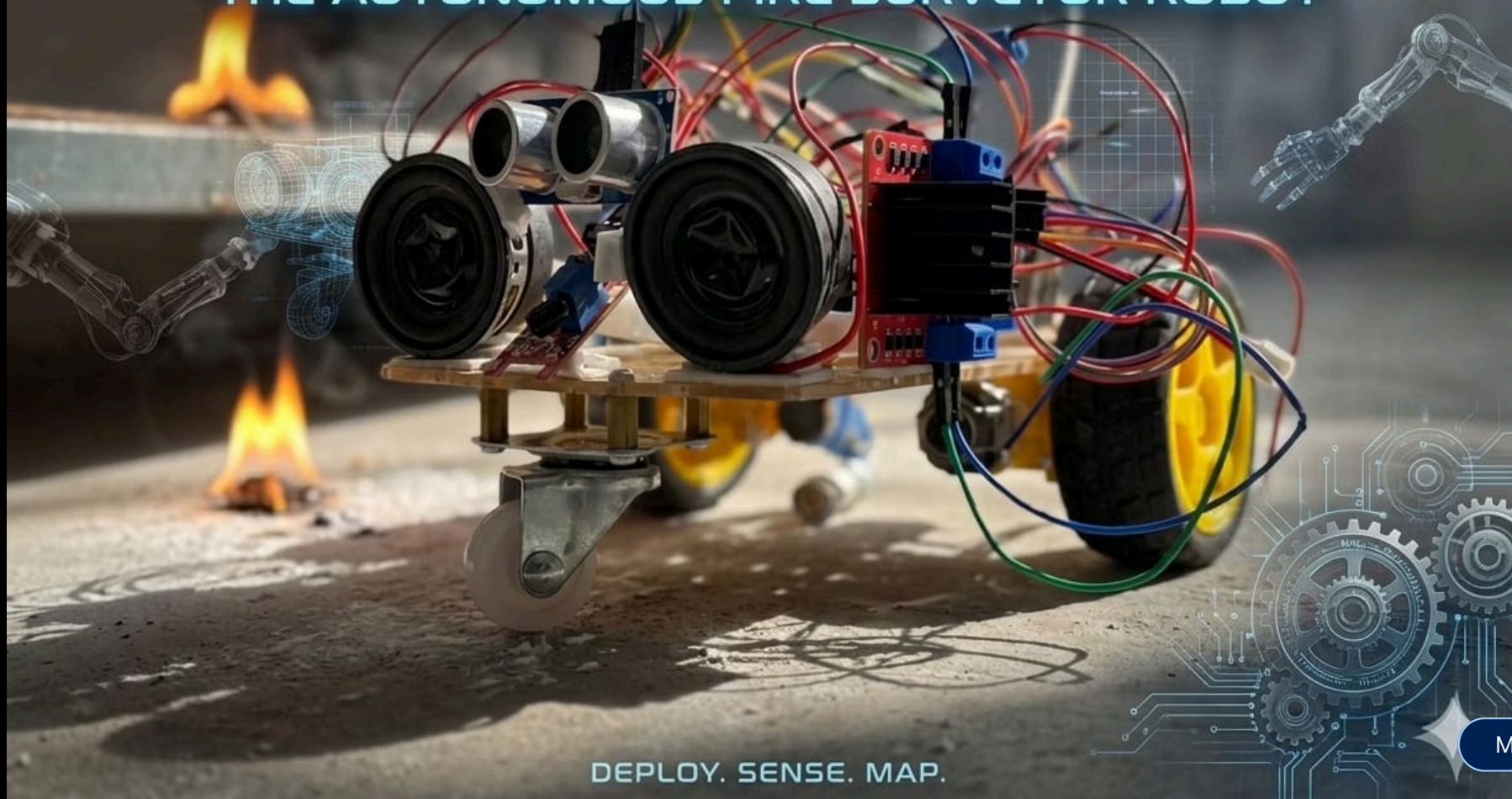


FLAMMEX

THE AUTONOMOUS FIRE SURVEYOR ROBOT



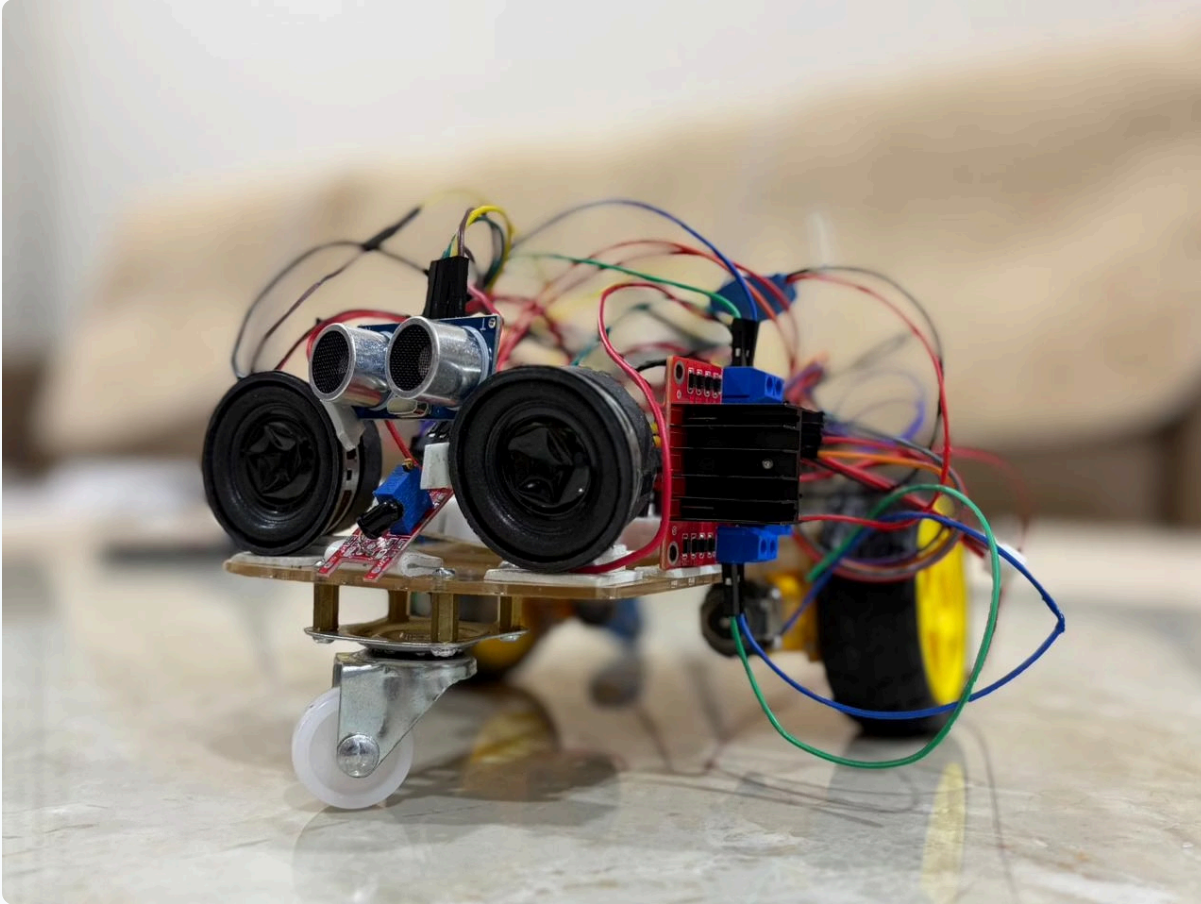
DEPLOY. SENSE. MAP.

Made with GAMMA

Team Members

- Shreeya Jadhav - Team Lead/Research and analysis
- Raghav Thakre - Technical Lead/System Design
- Bhargav Mhatre - Content Strategy/Communications
- Devesh Thakare - Documentation/Quality Assurance

Fire Fighting Robot



Autonomous robotic systems designed to detect, navigate, and suppress fires — protecting lives where humans cannot safely go.

The Problem

Firefighting remains one of the most dangerous professions. Extreme heat, toxic smoke, and structural collapse put first responders at severe risk every time they enter a burning building.



Our Solution



Autonomous Navigation

Moves through hazardous environments without human guidance.



Fire Detection

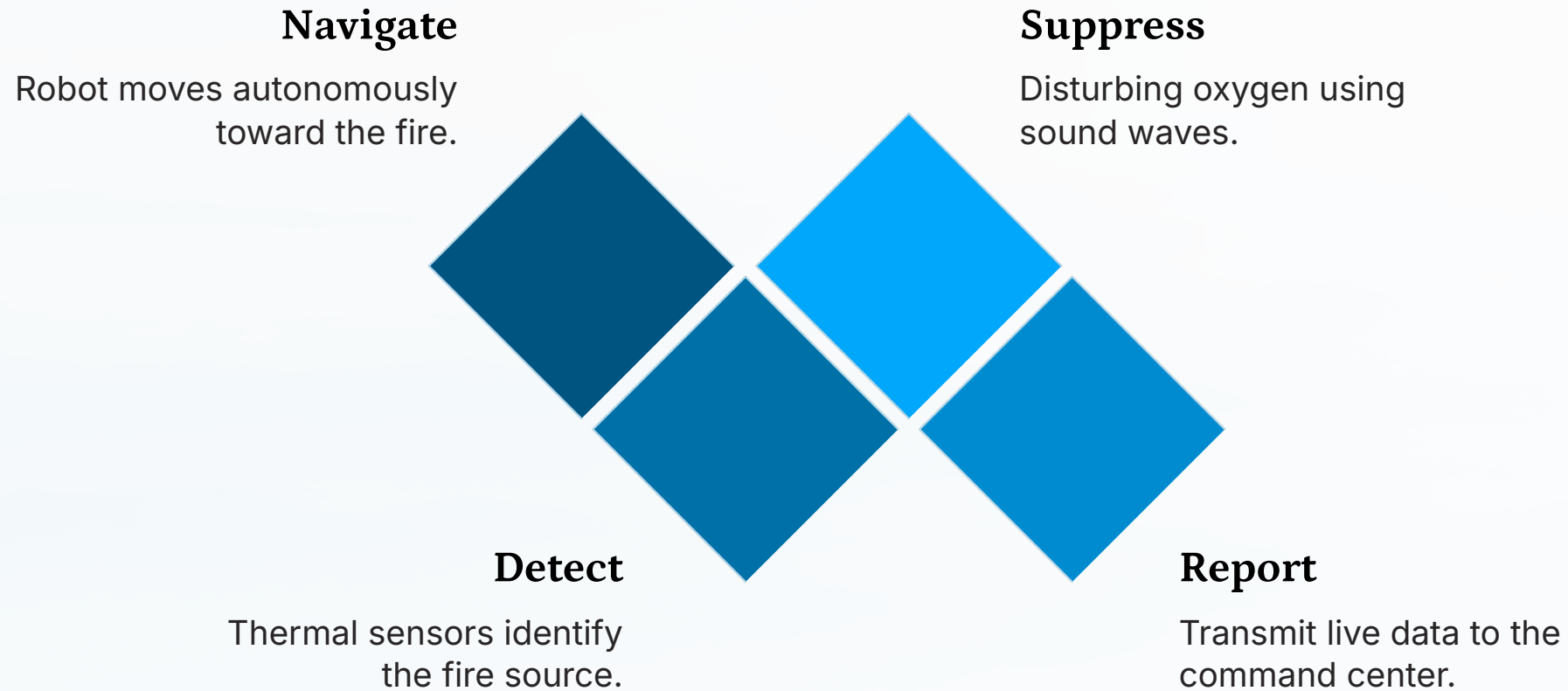
Infrared and thermal sensors locate heat sources instantly.



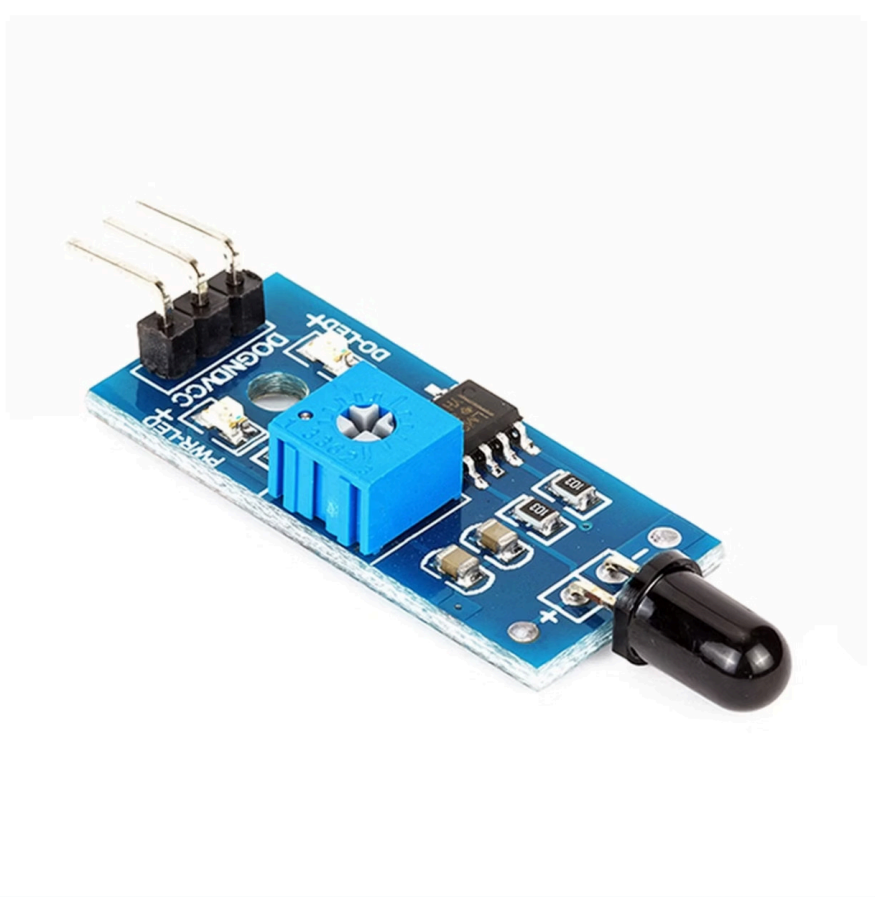
Suppression System

Low frequency sound waves extinguish flames precisely.

How It Works



The robot operates end-to-end autonomously — from detection to suppression — while streaming live data back to the command team.



Key Components

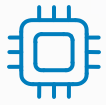
→ Flame Sensor

Detects the presence of fire by sensing infrared light emitted by flames.

→ Motorized Chassis

All-terrain wheels for navigating debris and uneven floors.

Electronics Components Explained



Arduino UNO

The microcontroller board that acts as the "brain" of the robot. It receives data from the sensors, processes it, and sends commands to the motors and speakers.



L298N Motor Driver

An intermediary board that acts as a switch. It allows the low-voltage Arduino to safely control the speed and spinning direction of the higher-voltage DC motors.



Castor wheel (support)

A free-swiveling, unpowered wheel typically placed at the front or back of the chassis to keep the two-wheeled robot balanced and stable, enhancing maneuverability.



Subwoofer / High Power Speaker

Outputs the amplified audio. In a fire-detecting robot, this is crucial for sounding a loud, clear alarm when a flame is spotted, alerting nearby personnel.



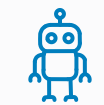
Jumper wires

Simple wires with pins at the ends used to easily connect all the electronic components and sensors together without needing to solder, allowing for flexible prototyping and assembly.



Ultrasonic Sensor (HC-SR04)

Acts as the robot's "eyes" for obstacle avoidance. It bounces sound waves off objects to measure distances, preventing collisions and allowing safe navigation.



2 × DC Motors + Wheels

The main drive system that provides mobility. By controlling each motor independently (differential steering), the robot can move forward, backward, and turn with precision.



Audio Amplifier Module (PAM8403 / TDA2030)

Takes low-power audio signals (like an alarm tone generated by the Arduino) and boosts them so they are strong enough to drive a larger speaker.



Battery (12V + 9V or Li-ion pack)

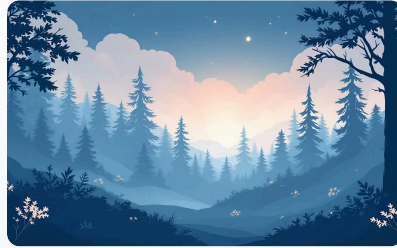
The main power source. A high-capacity pack is needed for power-hungry parts like motors and amplifiers, while the Arduino can be powered separately or via a regulated step-down.

Applications



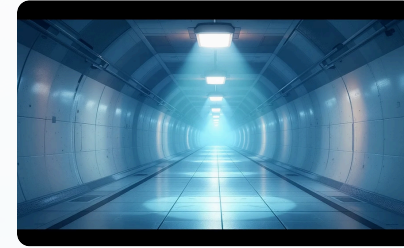
Industrial Facilities

Warehouses, factories, and chemical plants with high fire risk.



Wildfire Zones

Remote terrain where human access is dangerous or impossible.



Tunnels & Mines

Confined spaces with limited visibility and escape routes.



Technology Stack

Microcontroller

Arduino / Raspberry Pi for real-time processing and control.

Wireless Communication

RF and Wi-Fi modules for remote monitoring and override.

AI & Computer Vision

Machine learning models for flame and obstacle recognition.

Impact & Benefits

0

Human Lives at Risk

Robot enters the danger zone so firefighters don't have to.

24/7

Operational Readiness

Always on standby — no fatigue, no fear.

60s

Response Time

Rapid deployment from detection to active suppression.

Challenges & Future Scope

Current Challenges

Battery life, payload limits, and complex terrain navigation.

Future Enhancements

Swarm robotics, drone integration, and advanced AI decision-making.



The Future of Firefighting is Here

Fire fighting robots represent a transformative leap — combining autonomy, AI, and engineering to save lives, reduce risk, and redefine emergency response for the 21st century.



INNOVATION



SAFETY FIRST



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Conclusion

- The project successfully demonstrates a fire-fighting robot using sound waves.
- It uses sensors to detect fire and obstacles automatically.
- The Arduino controls movement and activates the fire suppression system.
- Low-frequency sound waves are generated using an amplifier and speakers.
- These sound waves disturb the oxygen supply required for combustion.
- The system is eco-friendly as it does not use water or chemicals.
- It is suitable for small-scale fire detection and suppression.
- The project highlights an innovative and modern approach to fire safety.
- However, its effectiveness is limited due to low power and small-scale components.
- The system can be improved using high-power speakers and advanced



Thank you

